



Universität Augsburg
Mathematisch-Naturwissenschaftlich-
Technische Fakultät

Classical networks for the Hubbard model with a tilted potential

- Master Thesis Colloquium -

Jonas Kell
Chair for theoretical Physics III

12th of March 2025

Outline

1 Introduction of the Physical Problem

2 Time Evolution

3 Sampling & Simplifications

4 Variational Classical Networks

5 Implementation & Experiments

6 Conclusion & Outlook

Particle Types

Introduction of the Physical Problem

- Bosonic operators
 - Commutation relations

$$\left[\hat{b}_l, \hat{b}_m \right] = 0$$

$$\left[\hat{b}_l^\dagger, \hat{b}_m^\dagger \right] = 0$$

$$\left[\hat{b}_l, \hat{b}_m^\dagger \right] = \delta_{l,m}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
 - Commutation relations
- Fermionic operators
 - Anti-commutation relations

$$\{\hat{c}_l, \hat{c}_m\} = 0$$

$$\{\hat{c}_l^\dagger, \hat{c}_m^\dagger\} = 0$$

$$\{\hat{c}_l, \hat{c}_m^\dagger\} = \delta_{l,m}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
 - Commutation relations
- Fermionic operators
 - Anti-commutation relations
- Hard-core bosonic operators
 - Commutation relations
 - Occupation number limited to 0 and 1
 - Number operator is idempotent

$$[\hat{h}_l, \hat{h}_m] = 0$$

$$[\hat{h}_l^\dagger, \hat{h}_m^\dagger] = 0$$

$$[\hat{h}_l, \hat{h}_m^\dagger] = \left(1 - 2 \cdot \hat{h}_m^\dagger \hat{h}_m\right) \cdot \delta_{l,m}$$

$$\left(\hat{h}_l^\dagger \hat{h}_l\right)^n = \hat{h}_l^\dagger \hat{h}_l \quad \forall n \in \mathbb{N}$$

Particle Types

Introduction of the Physical Problem

- Bosonic operators
 - Commutation relations
- Fermionic operators
 - Anti-commutation relations
- Hard-core bosonic operators
 - Commutation relations
 - Occupation number limited to 0 and 1
 - Number operator is idempotent
- Spins (Pauli operators/matrices)
 - Particles from different sites commute
 - Different spin directions on the same site anti-commute
 - They obey angular momentum commutation relations

$$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}^2 = \hat{\sigma}^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}^1 = \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}^3 = \hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\left\{ \hat{\sigma}_l^\alpha, \hat{\sigma}_l^\beta \right\} = 2 \cdot \delta_{\alpha,\beta}$$

$$\left[\hat{\sigma}_l^\alpha, \hat{\sigma}_m^\beta \right] = 2 \cdot i \cdot \varepsilon_{\alpha,\beta,\gamma} \cdot \delta_{l,m} \cdot \hat{\sigma}_l^\gamma$$

Particle Types - Mappings

Introduction of the Physical Problem

- Properties of Pauli matrices
 - Base of the complex 2x2 matrices
 - Can be directly mapped to hard-core bosons

$$\hat{\sigma}_l^x = \hat{h}_l + \hat{h}_l^\dagger$$

$$\hat{\sigma}_l^y = i \cdot (\hat{h}_l - \hat{h}_l^\dagger)$$

$$\hat{\sigma}_l^z = 2 \cdot \hat{h}_l^\dagger \hat{h}_l - 1$$

$$|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |0\rangle$$

$$\hat{\sigma}^+ |\downarrow\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |1\rangle = \hat{h}^\dagger |0\rangle$$

Hamiltonian of the system

Introduction of the Physical Problem

- Hubbard model Hamiltonian
- Constant external electric field
- Two particle types (spin degrees)
 - Different *spins* than for Pauli spin operators

$$\mathcal{H} = \mathcal{H}_0 + \hat{V}$$

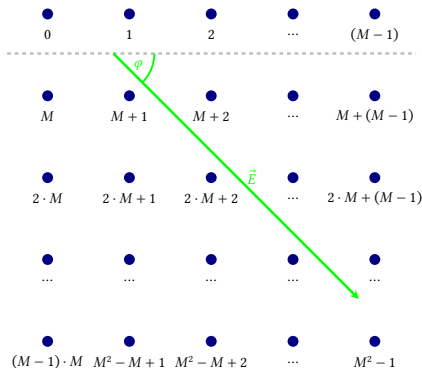
$$\mathcal{H}_0 = U \cdot \sum_l \hat{n}_{l,\uparrow} \hat{n}_{l,\downarrow} + \sum_{l,\sigma} \underbrace{\left(\vec{E} \cdot \vec{r}_l \right)}_{\varepsilon_l} \hat{n}_{l,\sigma}$$

$$\hat{V} = -J \cdot \sum_{\langle l,m \rangle, \sigma} \left(\hat{h}_{l,\sigma}^\dagger \hat{h}_{m,\sigma} + \hat{h}_{m,\sigma}^\dagger \hat{h}_{l,\sigma} \right)$$

Geometry of the system

Introduction of the Physical Problem

- 2-dimensional geometry
- Square lattice arrangement
- Computation for general size and field angle



Goal of the Calculation

Time Evolution

- Evaluate the time-evolution of the system for various observables
 - Uses calculation in the *Interaction Picture*
 - Introduce operator-less *effective Hamiltonian*

$$|\psi^S(t)\rangle = \sum_N e^{H_{\text{eff}}(N,t)} \psi_N |N\rangle$$

The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N, t) = -iE_0(N)t + \mathcal{H}_N(t)$$

- Constructed from the contributions of the base energy and the perturbation
- Base energy contribution:

$$\begin{aligned} E_0(N) &= \frac{\langle N | \mathcal{H}_0 | N \rangle}{\langle N | N \rangle} \\ &= \langle N | U \cdot \sum_l \hat{n}_{l,\uparrow} \hat{n}_{l,\downarrow} | N \rangle + \langle N | \sum_{l,\sigma} \varepsilon_l \hat{n}_{l,\sigma} | N \rangle \\ &= U \cdot \sum_l n_{l,\uparrow} n_{l,\downarrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} \end{aligned}$$

The effective Hamiltonian

Time Evolution

- To evaluate the time-evolution of the perturbation

$$\hat{V} = -J \cdot \sum_{\langle l,m \rangle} \left(\hat{h}_l^\dagger \hat{h}_m + \hat{h}_m^\dagger \hat{h}_l + \hat{d}_l^\dagger \hat{d}_m + \hat{d}_m^\dagger \hat{d}_l \right)$$

- Solve the equation of motion for the ladder operators

$$\begin{aligned}\hat{h}_m^{\dagger I}(t) &= e^{i\varepsilon_m t} \left(1 + (e^{iU \cdot t} - 1) \hat{d}_m^{\dagger S} \hat{d}_m^S \right) \hat{h}_m^{\dagger S} \\ \hat{h}_m^I(t) &= e^{-i\varepsilon_m t} \left(1 + (e^{-iU \cdot t} - 1) \hat{d}_m^{\dagger S} \hat{d}_m^S \right) \hat{h}_m^S \\ \hat{d}_m^{\dagger I}(t) &= e^{i\varepsilon_m t} \left(1 + (e^{iU \cdot t} - 1) \hat{h}_m^{\dagger S} \hat{h}_m^S \right) \hat{d}_m^{\dagger S} \\ \hat{d}_m^I(t) &= e^{-i\varepsilon_m t} \left(1 + (e^{-iU \cdot t} - 1) \hat{h}_m^{\dagger S} \hat{h}_m^S \right) \hat{d}_m^S\end{aligned}$$

The effective Hamiltonian

Time Evolution

- Insert and reorder the operators

$$\begin{aligned}\hat{V}^I(t) &= \left\{ \hat{V}^S \right\}^I(t) = -J \cdot \sum_{[l,m]} \left\{ \left(\hat{h}_l^{\dagger S} \hat{h}_m^S + \hat{d}_l^{\dagger S} \hat{d}_m^S \right) \right\}^I(t) \\&= -J \cdot \sum_{[l,m]} \left(\hat{h}_l^{\dagger I}(t) \hat{h}_m^I(t) + \hat{d}_l^{\dagger I}(t) \hat{d}_m^I(t) \right) \\&= -J \cdot \sum_{[l,m]} \left[\Lambda_A(l, m, t) \cdot \hat{F}_A(l, m) + \Lambda_B(l, m, t) \cdot \hat{F}_B(l, m) + \Lambda_C(l, m, t) \cdot \hat{F}_C(l, m) \right] \\ \Lambda_A(l, m, t) &= e^{i(\varepsilon_l - \varepsilon_m) \cdot t} & \hat{F}_A(l, m) &= \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{h}_{l,\sigma}^{\dagger S} \hat{h}_{m,\sigma}^S \left(1 + 2 \cdot \hat{n}_{l,\bar{\sigma}}^S \hat{n}_{m,\bar{\sigma}}^S - \hat{n}_{l,\bar{\sigma}}^S - \hat{n}_{m,\bar{\sigma}}^S \right) \\ \Lambda_B(l, m, t) &= e^{i(\varepsilon_l - \varepsilon_m + U) \cdot t} & \hat{F}_B(l, m) &= \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{h}_{l,\sigma}^{\dagger S} \hat{h}_{m,\sigma}^S \left(\hat{n}_{l,\bar{\sigma}}^S - \hat{n}_{l,\bar{\sigma}}^S \hat{n}_{m,\bar{\sigma}}^S \right) \\ \Lambda_C(l, m, t) &= e^{i(\varepsilon_l - \varepsilon_m - U) \cdot t} & \hat{F}_C(l, m) &= \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{h}_{l,\sigma}^{\dagger S} \hat{h}_{m,\sigma}^S \left(\hat{n}_{m,\bar{\sigma}}^S - \hat{n}_{m,\bar{\sigma}}^S \hat{n}_{l,\bar{\sigma}}^S \right)\end{aligned}$$

The effective Hamiltonian

Time Evolution

- Evaluate the contribution to the effective Hamiltonian
 - Requires previously calculated value of the V-operator in the Interaction Picture
 - Controllable *cumulant expansion*

$$\begin{aligned}\mathcal{H}_N(t) &= \sum_{v=1}^{\infty} \frac{(-i)^v}{v!} \int_0^t dt_1 \int_0^t dt_2 \cdots \int_0^t dt_v \langle \mathbb{T} \hat{V}^I(t_1) \hat{V}^I(t_2) \cdots \hat{V}^I(t_v) \rangle_{c(N)} \\ &= -i \int_0^t dt_1 \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} \\ &\quad - \frac{1}{2} \int_0^t dt_1 \int_0^t dt_2 \left(\frac{\langle N | \mathbb{T} \hat{V}^I(t_1) \hat{V}^I(t_2) | \Psi^S \rangle}{\langle N | \Psi^S \rangle} - \frac{\langle N | \hat{V}^I(t_1) | \Psi^S \rangle \cdot \langle N | \hat{V}^I(t_2) | \Psi^S \rangle}{\langle N | \Psi^S \rangle^2} \right) \\ &\quad + \cdots\end{aligned}$$

Handling of Observables

Time Evolution

- This allows for general evaluation of expectation values
 - Requires a probability for the state
 - Requires a local observable

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} = \sum_N P(N, t) \underbrace{\sum_K \langle N | \hat{\mathcal{O}} | K \rangle e^{\mathcal{H}_{\text{eff}}(K, t) - \mathcal{H}_{\text{eff}}(N, t)} \frac{\Psi_K}{\Psi_N}}_{\hat{\mathcal{O}}_{\text{loc}}(N, t)} = \sum_N P(N, t) \hat{\mathcal{O}}_{\text{loc}}(N, t)$$

Simple Observables

Time Evolution

- Local observable for double occupation measurement
 - Observables generally are very sparse matrices
 - Reduces to pure occupation measurement

$$\hat{\mathcal{O}}_{\text{loc}}(N, t) = \sum_K \langle N | \hat{\mathcal{O}}_{\text{do}}(l) | K \rangle e^{\mathcal{H}_{\text{eff}}(K, t) - \mathcal{H}_{\text{eff}}(N, t)} \frac{\Psi_K}{\Psi_N} = n_{l, \uparrow} \cdot n_{l, \downarrow}$$

- Local observable for particle current measurement
 - Requires evaluation of the difference of two effective Hamiltonians

$$\hat{\mathcal{O}}_{\text{loc}}(N, t) = i \cdot \left[n_{m, \sigma} \cdot (1 - n_{l, \sigma}) - n_{l, \sigma} \cdot (1 - n_{m, \sigma}) \right] \cdot \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N}, t) - \mathcal{H}_{\text{eff}}(N, t)}$$

Access to Density-Matrices

Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix
 - Access to *purity*, *concurrence* and other entanglement measures/monotones
- Direct calculation not possible

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_{k=0}^M (\mathbb{1}_A \otimes \langle \chi_k |) \rho_{AB} (\mathbb{1}_A \otimes |\chi_k\rangle)$$

- Use Pauli matrices to expand the complex 4x4 matrix

$$\rho_A(t) = \rho_{l,\sigma,m,\mu}(t) = \frac{1}{4} \sum_{\alpha,\beta \in \{0,x,y,z\}} \langle \hat{\sigma}_{l,\sigma}^\alpha \hat{\sigma}_{m,\mu}^\beta \rangle_{(t)} (\hat{\sigma}^\alpha \otimes \hat{\sigma}^\beta)$$

Access to Density-Matrices

Time Evolution

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^x \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)}$$

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^y \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{l,\sigma})$$

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^z \hat{\sigma}_{m,\mu}^0 : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot (2 \cdot n_{l,\sigma} - 1)$$

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^x \hat{\sigma}_{m,\mu}^x : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t) - \mathcal{H}_{\text{eff}}(N,t)}$$

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^y \hat{\sigma}_{m,\mu}^y : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot (2 \cdot n_{l,\sigma} - 1) \cdot (1 - 2 \cdot n_{m,\mu})$$

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^x \hat{\sigma}_{m,\mu}^y : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{m,\mu})$$

$$\hat{\mathcal{O}} = \hat{\sigma}_{l,\sigma}^y \hat{\sigma}_{m,\mu}^x : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t) - \mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{l,\sigma})$$

Probability Calculations

Sampling & Simplifications

- Full probability still requires normalization
- Switch to the Metropolis *Hastings* algorithm
 - Non-exponential number of samples
 - Depends only on a transition probability
 - Does not need normalization because it cancels

$$\begin{aligned}\alpha &= \frac{P(\tilde{N}, t)}{P(N, t)} = \frac{f(\tilde{N}, t)}{f(N, t)} = \frac{|e^{\mathcal{H}_{\text{eff}}(\tilde{N}, t)}|^2 |\Psi_{\tilde{N}}|^2}{|e^{\mathcal{H}_{\text{eff}}(N, t)}|^2 |\Psi_N|^2} \\&= \frac{|\Psi_{\tilde{N}}|^2 e^{\Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) + i\Im(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) + \Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) - i\Im(\mathcal{H}_{\text{eff}}(\tilde{N}, t))}}{|\Psi_N|^2 e^{\Re(\mathcal{H}_{\text{eff}}(N, t)) + i\Im(\mathcal{H}_{\text{eff}}(N, t)) + \Re(\mathcal{H}_{\text{eff}}(N, t)) - i\Im(\mathcal{H}_{\text{eff}}(N, t))}} \\&= \frac{|\Psi_{\tilde{N}}|^2}{|\Psi_N|^2} e^{2\Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t)) - 2\Re(\mathcal{H}_{\text{eff}}(N, t))} \\&= \frac{|\Psi_{\tilde{N}}|^2}{|\Psi_N|^2} e^{2\Re(\mathcal{H}_{\text{eff}}(\tilde{N}, t) - \mathcal{H}_{\text{eff}}(N, t))}\end{aligned}$$

$$\frac{\langle \Psi^S(t) | \hat{\mathcal{O}} | \Psi^S(t) \rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} = \sum_N P(N, t) \cdot \hat{\mathcal{O}}_{\text{loc}}(N, t) \approx \frac{1}{|\{N\}_{\text{MC}}|} \sum_{\{N\}_{\text{MC}}} \hat{\mathcal{O}}_{\text{loc}}(N, t)$$

Analytical Simplifications

Sampling & Simplifications

- Choose a helpful initial state

$$\Psi_N = \frac{1}{\sqrt{\#(\text{states})}} = \frac{1}{\sqrt{2^{\#(\text{sites})} \cdot 2}} = \frac{1}{2^{\#(\text{sites})}}$$

$$|\Psi^S(t=0)\rangle = \bigotimes_{l=1}^{\#(\text{sites})} \frac{1}{2} \left(1 + \hat{h}_{l,\uparrow}^{\dagger S} + \hat{h}_{l,\downarrow}^{\dagger S} + \hat{h}_{l,\uparrow}^{\dagger S} \hat{h}_{l,\downarrow}^{\dagger S} \right) |0\rangle$$

- Now *almost all* terms cancel in *differences* of the effective Hamiltonian
 - Exemplary for single flip on base energy term

$$\begin{aligned} E_0(N) - E_0(\tilde{N}) &= U \sum_l n_{l,\downarrow} n_{l,\uparrow} - U \sum_l \tilde{n}_{l,\downarrow} \tilde{n}_{l,\uparrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} - \sum_{l,\sigma} \varepsilon_l \tilde{n}_{l,\sigma} \\ &= \varepsilon_i (2n_{i,\sigma_i} - 1) + U \cdot \begin{cases} \sigma_i = \uparrow : & n_{i,\downarrow} (2n_{i,\uparrow} - 1) \\ \sigma_i = \downarrow : & n_{i,\uparrow} (2n_{i,\downarrow} - 1) \end{cases} \end{aligned}$$

Why and How should Parameters be variational?

Variational Classical Networks

- Theoretically: always better approximation through higher order terms
 - Hard to calculate analytically and expensive to evaluate
- Hope: Better behavior if parameters are *learned/optimized*, starting from the analytical result

$$|\Psi_{\vec{\eta}+\delta\cdot\dot{\vec{\eta}}}(t+\delta)\rangle \leftrightarrow e^{-iH\delta} |\Psi_{\vec{\eta}}(t)\rangle$$

$$\vec{\eta}(t+\delta) = \vec{\eta}(t) + \delta \cdot \dot{\vec{\eta}}(t)$$

$$O_k(N) = \frac{\partial \ln \Psi_{N,\vec{\eta}}(t)}{\partial \eta_k} = \frac{\partial \ln}{\partial \eta_k} e^{\mathcal{H}_{\text{eff}}(N,\vec{\eta},t) + \ln \Psi_N}$$

$$E_{\text{loc}}(N) = \frac{\langle N | \mathcal{H} | \Psi_{\vec{\eta}}(t) \rangle}{\langle N | \Psi_{\vec{\eta}}(t) \rangle}$$

Why and How should Parameters be variational?

Variational Classical Networks

- Theoretically: always better approximation through higher order terms
 - Hard to calculate analytically and expensive to evaluate
- Hope: Better behavior if parameters are *learned/optimized*, starting from the analytical result

$$|\Psi_{\vec{\eta}+\delta\cdot\dot{\vec{\eta}}}(t+\delta)\rangle \leftrightarrow e^{-iH\delta} |\Psi_{\vec{\eta}}(t)\rangle$$

$$\vec{\eta}(t+\delta) = \vec{\eta}(t) + \delta \cdot \dot{\vec{\eta}}(t)$$

$$\vec{S}_{k,k'} = \langle O_k^* O_{k'} \rangle_{(\vec{\eta})} - \langle O_k^* \rangle_{(\vec{\eta})} \cdot \langle O_{k'} \rangle_{(\vec{\eta})}$$

$$\vec{F}_k = \langle E_{\text{loc}} O_k^* \rangle_{(\vec{\eta})} - \langle E_{\text{loc}} \rangle_{(\vec{\eta})} \cdot \langle O_k^* \rangle_{(\vec{\eta})}$$

$$\sum_{k'} \vec{S}_{k,k'} \dot{\vec{\eta}}_{k'} = -i \vec{F}_k$$

$$\dot{\vec{\eta}} = -i \overleftrightarrow{S}^{-1} \vec{F}$$

First Try: Cumulant Expansion Prefactors

Variational Classical Networks

- Idea: generate a variational Hamiltonian by replacing the analytical prefactors

$$\mathcal{H}_N(t) = \sum_i C_i(t) \cdot \Phi_i(N) \quad \leftrightarrow \quad \mathcal{H}_{\text{VCN}}(N, \vec{\eta}) = \sum_i \eta_i \cdot \Phi_i(N)$$

$C_1(t) = \Pi_A(0, 1, t)$	$\Phi_1(N) = J \sum_l \sum_{\langle l < m \rangle^{+(x)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_A(l, m) K \rangle$	$C_5(t) = \Pi_B(1, 0, t)$	$\Phi_5(N) = J \sum_l \sum_{\langle l > m \rangle^{+(x)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_B(l, m) K \rangle$
$C_2(t) = \Pi_B(0, 1, t)$	$\Phi_2(N) = J \sum_l \sum_{\langle l < m \rangle^{+(x)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_B(l, m) K \rangle$	$C_6(t) = \Pi_C(1, 0, t)$	$\Phi_6(N) = J \sum_l \sum_{\langle l > m \rangle^{+(x)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_C(l, m) K \rangle$
$C_3(t) = \Pi_C(0, 1, t)$	$\Phi_3(N) = J \sum_l \sum_{\langle l < m \rangle^{+(x)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_C(l, m) K \rangle$	$C_7(t) = \Pi_A(0, M, t)$	$\Phi_7(N) = J \sum_l \sum_{\langle l < m \rangle^{+(y)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_A(l, m) K \rangle$
$C_4(t) = \Pi_A(1, 0, t)$	$\Phi_4(N) = J \sum_l \sum_{\langle l > m \rangle^{+(x)}} \sum_K \frac{\Psi_K}{\Psi_N} \langle N \hat{F}_A(l, m) K \rangle$	...	...

Correction: Watch Explicit Time-Dependency

Variational Classical Networks

- First strategy is not suitable
 - Energy- and variance is not conserved
- Problem: explicit time-dependency of base energy

$$\frac{\partial \mathcal{H}_{\text{eff}}(N, \vec{\eta}, t)}{\partial \vec{\eta}_k} = \frac{\partial (-iE_0(N)t + \mathcal{H}_{\text{VCN}}(N, \vec{\eta}))}{\partial \vec{\eta}_k}$$

- Solution: Replace base energy factors with variational parameters

$$E_0(N) = U \cdot \sum_m n_{m,\uparrow} n_{m,\downarrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma}$$
$$\eta_l^{\text{b.e.}} = \begin{cases} l = 0 : & -i \cdot U \cdot t \\ l > 0 : & -i \cdot \varepsilon_l \cdot t \end{cases}$$
$$\Phi_l^{\text{b.e.}}(N) = \begin{cases} l = 0 : & \sum_m n_{m,\uparrow} n_{m,\downarrow} \\ l > 0 : & \sum_{\sigma} n_{l,\sigma} \end{cases}$$
$$\mathcal{H}_{\text{eff}}^{\text{VCN}}(N, \vec{\eta}) = \sum_{n=1}^{\#(\text{var.})} \eta_n^{\text{var.}} \cdot \Phi_n^{\text{var.}}(N) + \sum_{l=0}^L \eta_l^{\text{b.e.}} \cdot \Phi_l^{\text{b.e.}}(N) = \sum_l \eta_l \cdot \Phi_l(N)$$
$$\vec{\eta} = (\eta_1^{\text{var.}}, \eta_2^{\text{var.}}, \dots, \eta_{\#(\text{var.})}^{\text{var.}}, \eta_0^{\text{b.e.}}, \eta_1^{\text{b.e.}}, \dots, \eta_L^{\text{b.e.}})^T$$

Implementation

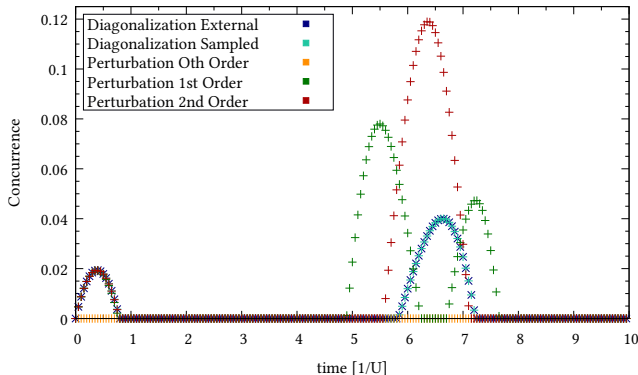
Implementation & Experiments

- Too much code to properly show
 - Look at repository for overview
- Testing: validation code to check the correct implementation
 - Independent implementations for same functionality
 - Sanity checks for state of data and program
- Optimization: runtime comparison for different code styles
 - For generated code fast execution over readability

Numerical Experiments

Implementation & Experiments

- Sadly no presentation possible in the small time allocated for the presentation
- The plots are in the presentation in the supplementary material



Conclusion & Outlook

Conclusion & Outlook

Conclusion:

- Successfully transferred and validated the theory
- Working implementation
 - Extensively tested
 - Configurable and extensible
- Started validating expectations with measurements on the compute cluster

Outlook:

- Re-write in lower level language to gain performance over Python
- Start more real experiments now that the theory is validated to work

Thank you for your kind attention



Universität Augsburg
Mathematisch-Naturwissenschaftlich-
Technische Fakultät

Jonas Kell
University of Augsburg
jonas.kell@student.uni-augsburg.de
www.uni-augsburg.de

References I

Summary & Conclusion

- [1] J. Kell, *Latex sources for this presentation and the main thesis*, (2025)
<https://github.com/jonas-kell/master-thesis-documents>.
- [2] J. Kell, *Thesis code implementation and reference*, (2025)
<https://github.com/jonas-kell/master-thesis-code>.
- [3] J. Kell, *Https://github.com/jonas-kell/beamer-theme-unia-mntf*, (2024)
<https://github.com/jonas-kell/math-manipulator>.

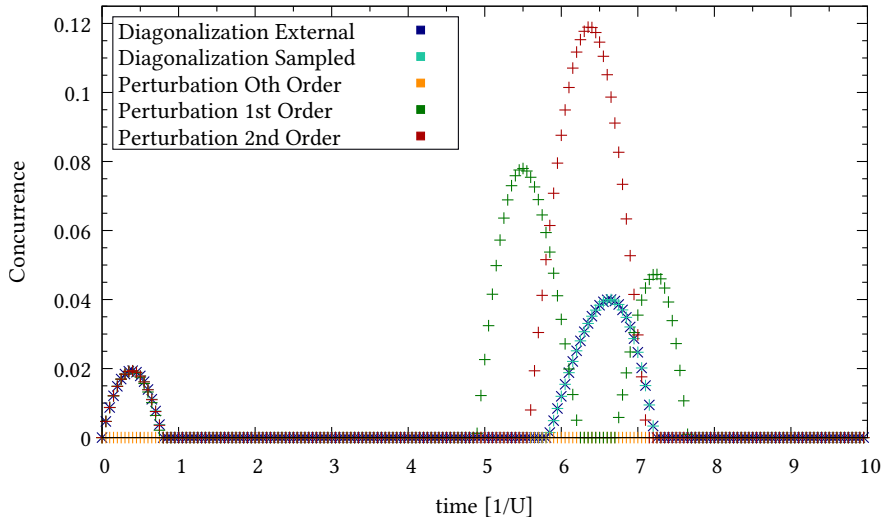
Extra Plots: Overview

Extra slides

- Exact comparison density matrix
- First orders: energy and variance comparison
- J parameter sweep
- Monte Carlo sampling convergence
- Runtime validation
- VCN: parametrization failure
- VCN: energy conservation can be controlled
- VCN: minimal square reference system
- VCN: larger square reference system in direct comparison
- VCN: system size dependency

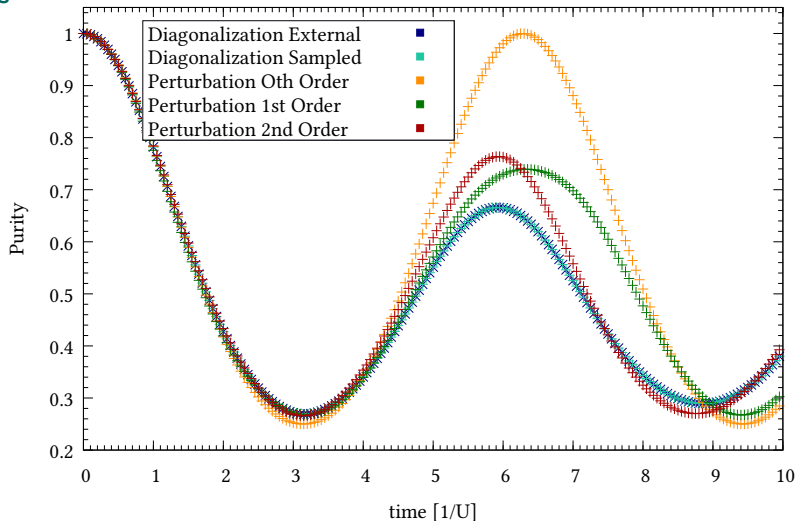
Exact Comparison: Concurrence

Extra slides



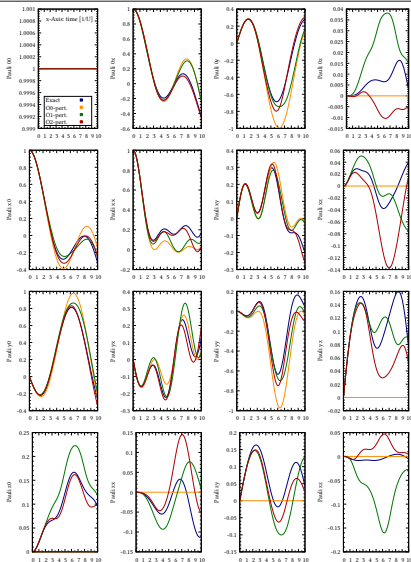
Exact Comparison: Purity

Extra slides



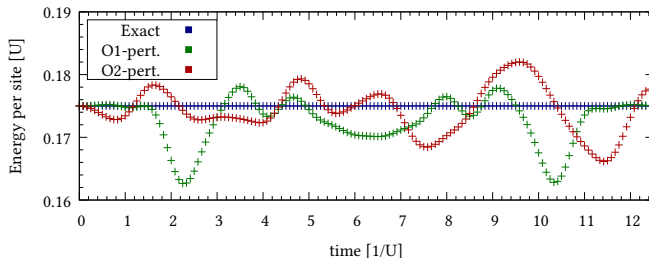
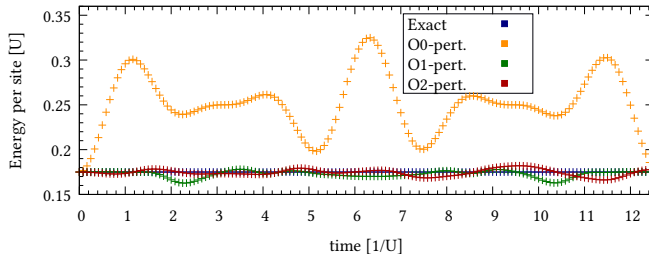
Exact Comparison: Pauli Measurements

Extra slides



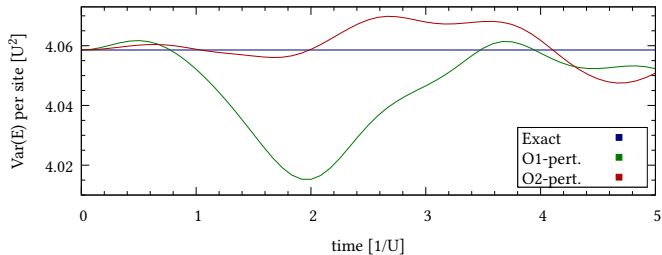
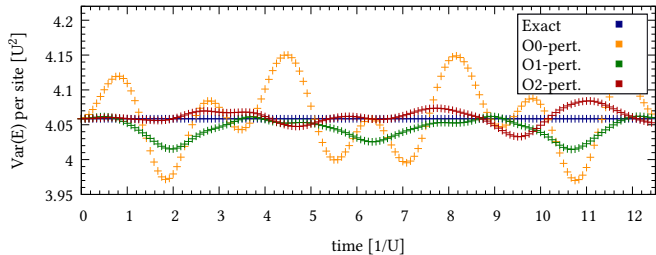
First Orders: Energy Comparison

Extra slides



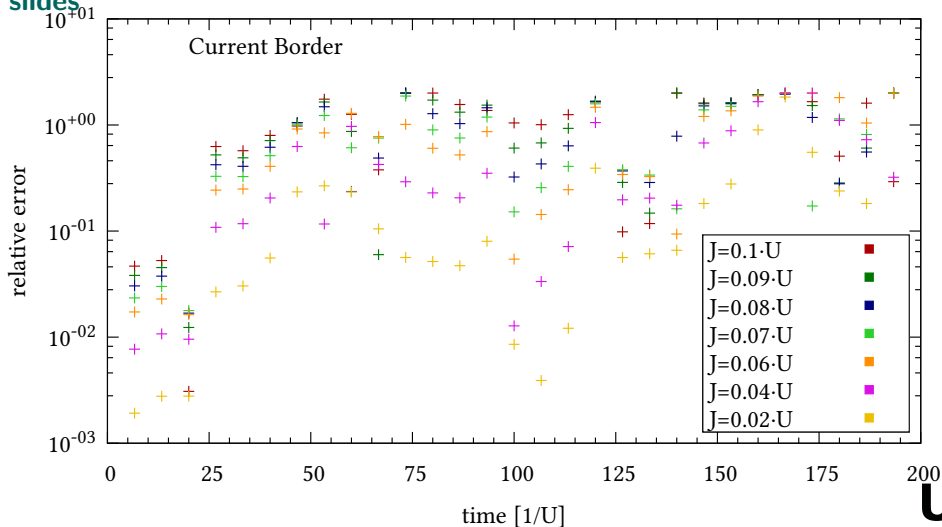
First Orders: Variance Comparison

Extra slides



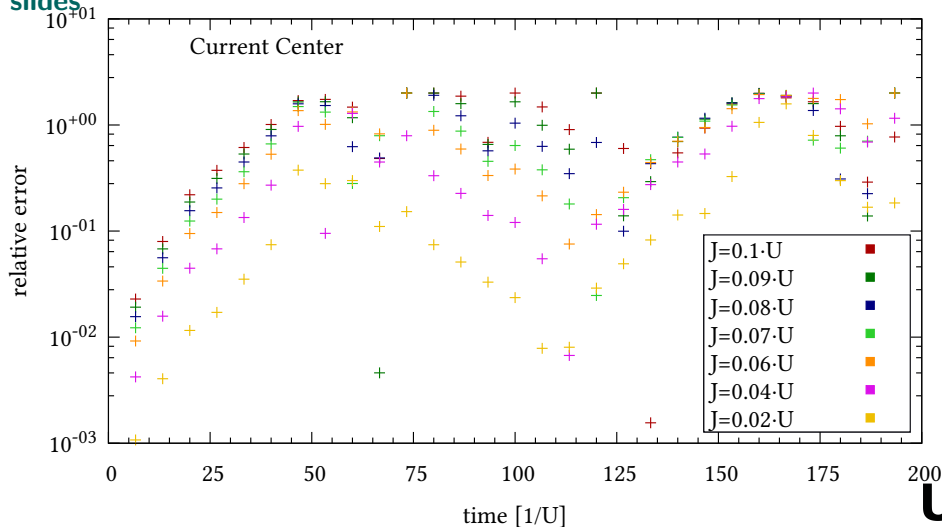
J-Sweep: Current Border

Extra slides



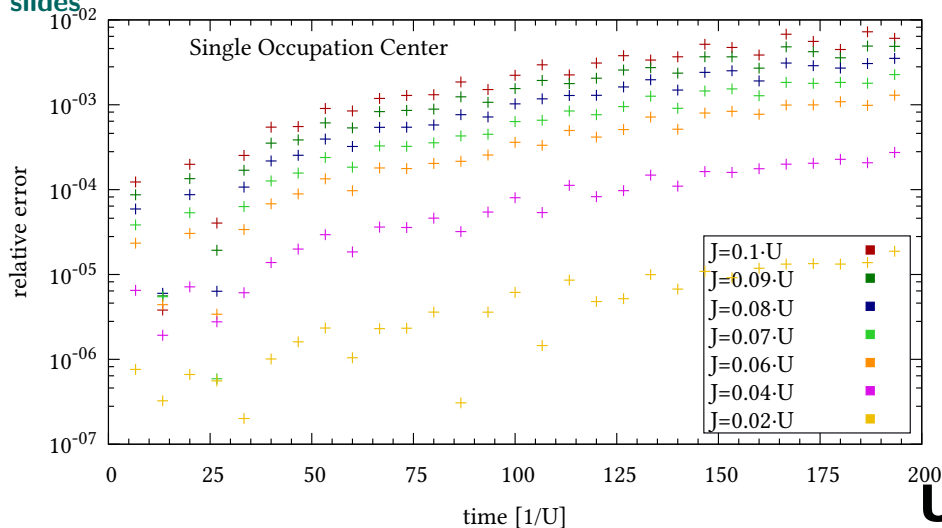
J-Sweep: Current Center

Extra slides



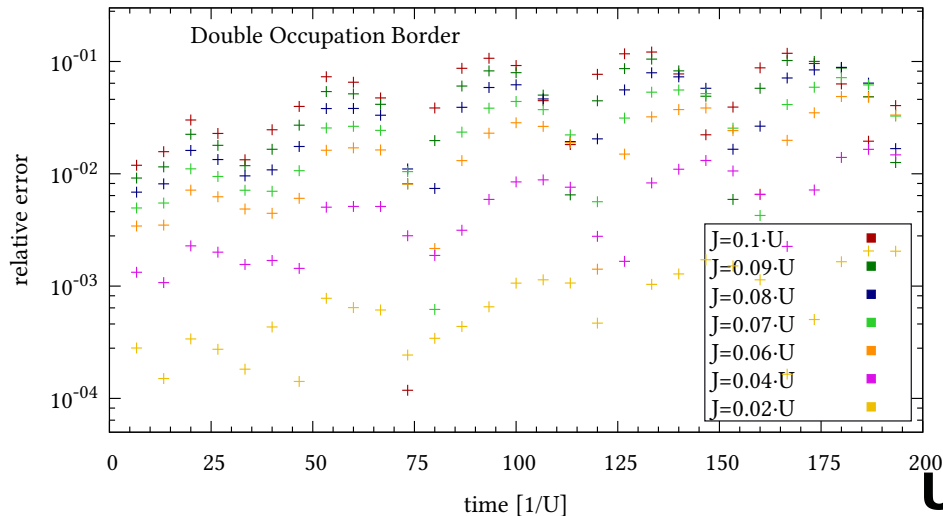
J-Sweep: Single Occupation Center

Extra slides



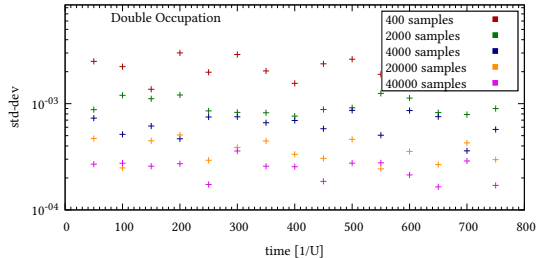
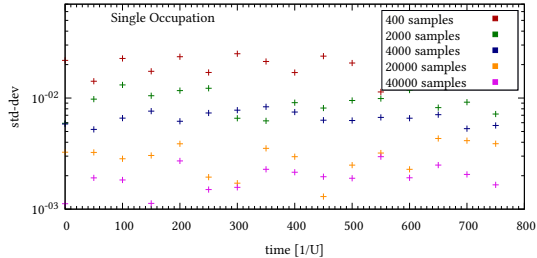
J-Sweep: Double Occupation Border

Extra slides



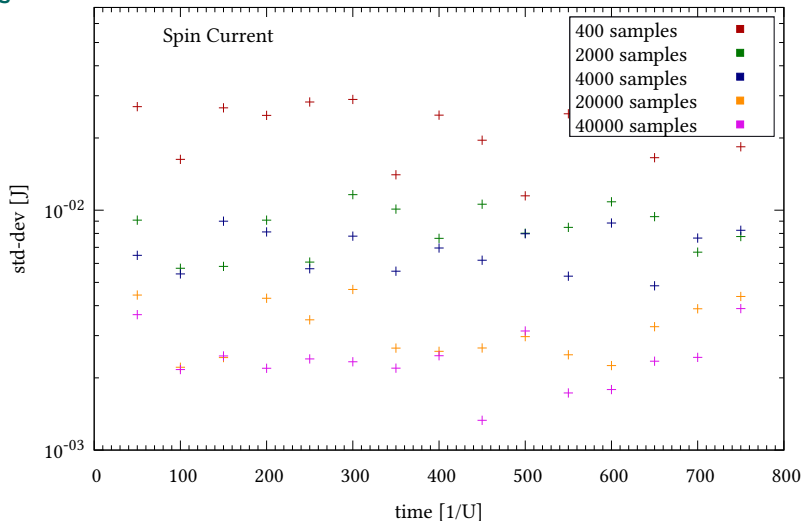
Monte Carlo Convergence: Occupation

Extra slides



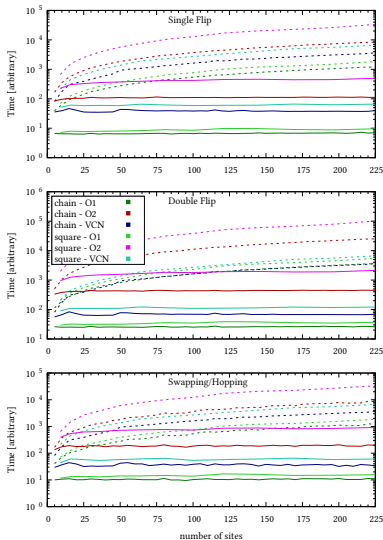
Monte Carlo Convergence: Occupation

Extra slides



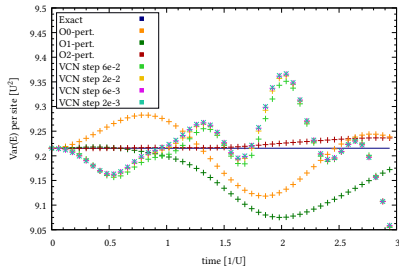
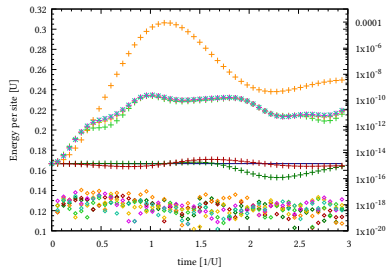
Runtime Validation

Extra slides



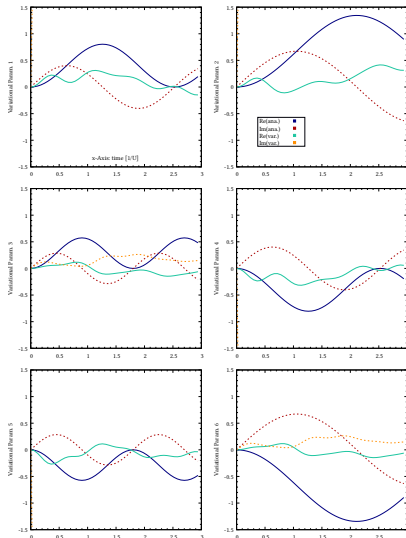
VCN Failure: Non-Explicit Parametrization

Extra slides



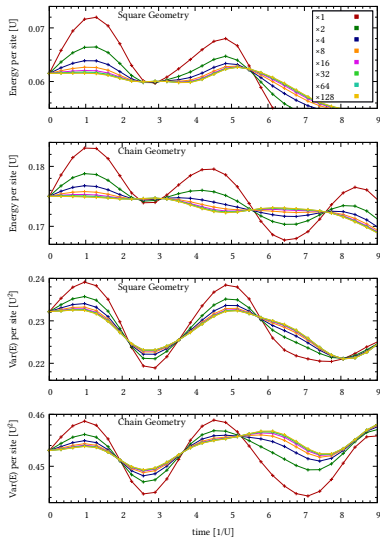
VCN Failure: Non-Explicit Parametrization

Extra slides



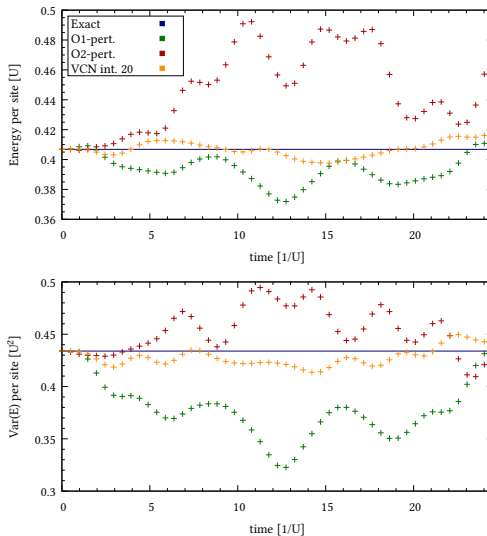
VCN Validation: Energy Conservation Controllable

Extra slides



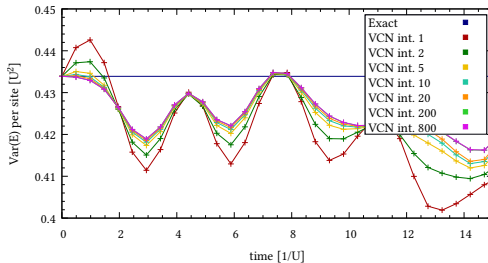
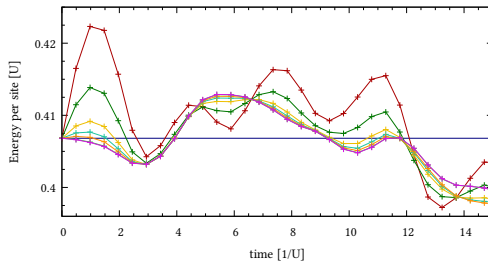
VCN: Small Square System

Extra slides



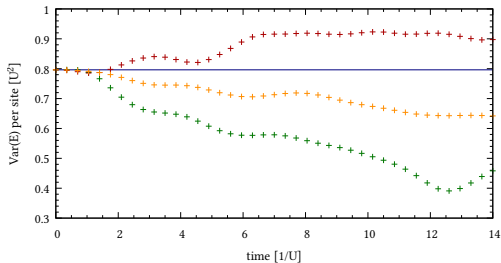
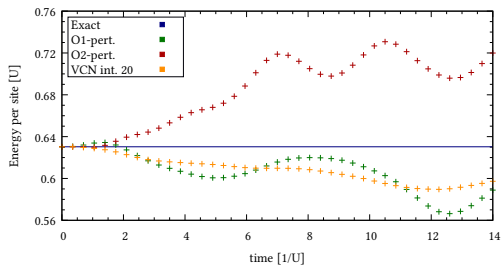
VCN: Small Square System

Extra slides



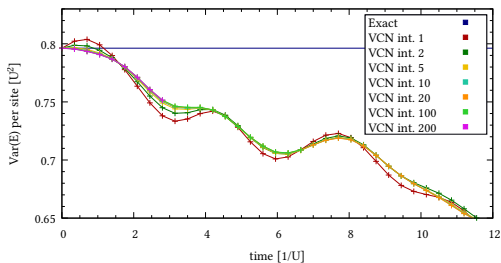
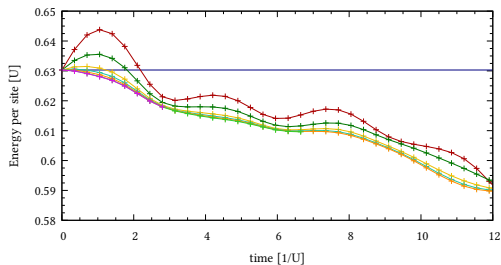
VCN: Bigger Square System

Extra slides



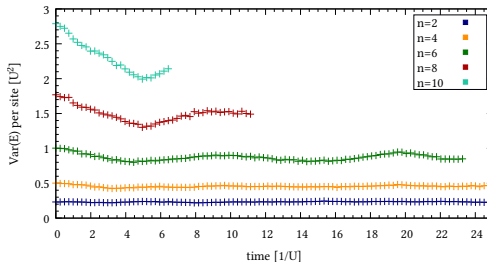
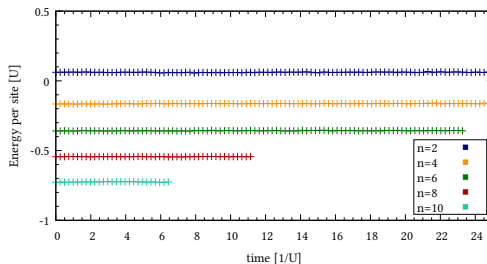
VCN: Bigger Square System

Extra slides



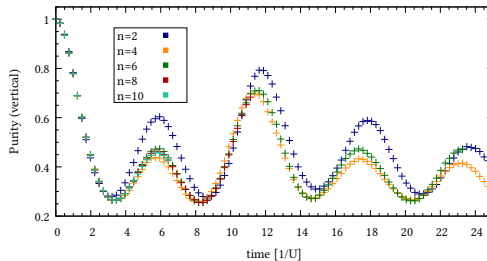
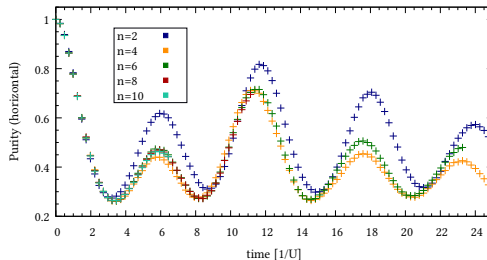
VCN End to End Test: System Size Dependency

Extra slides



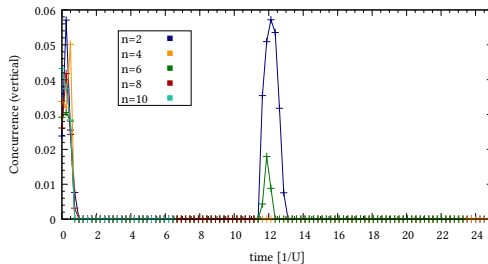
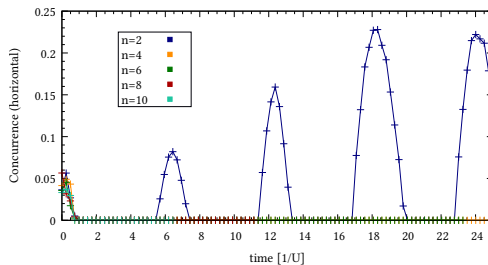
VCN End to End Test: System Size Dependency

Extra slides



VCN End to End Test: System Size Dependency

Extra slides



VCN End to End Test: System Size Dependency

Extra slides

